JOHN (IOANNIS) MARIS

Data Science, MSc.

About Me

Belgium, Brussels, Evere

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Portfolio

LinkedIn

GitHub

Honors & Awards

APPLE MSc Scholarship - Disentangled Representation Learning via Mutual Information Optimization.
Awarded from Apple, through IACM FORTH for my research in protein engineering to develop enzymes for plastic degradation.(Nov. 2024).

Erasmus+ Traineeship Grant. 3 months
Awarded from Technical University of Crete.

Erasmus+ Traineeship Grant. 4 months
Awarded from University of Crete.

Interests

- Machine Learning
- Statistics & Causality
- Deep Generative AI
- Natural Language Processing
- Bioinformatics
- Time Series & Econometrics
- Automotive Engineering
- Dynamical Systems

Language

- English (Michigan)
- Greek (Native)
- Japanese (Elementary)

Soft Skills

- Time Management
- Teamwork
- Problem Solving

Education

Master of Science in Data Analysis & Machine-Statistical Learning.

Oct. 2023 - Jul. 2025

110 ECTS programme | GPA: 9.0 (Excellent). | Supervisor: Yiannis Pantazis.
Thesis topic: Generative AI in Protein Engineering using Large Language Diffusion Models.
Organizing bodies:
University of Crete: Dep. of Mathematics and Applied Mathematics & Dep. of Computer Science; Foundation of Research & Technology Hellas (FORTH): Institute of Applied and Computational Mathematics (IACM) & Institute of Computer Science

Bachelor of Science in Mathematics & Applied Mathematics.

Oct. 2017 - Sep. 2022

274/240 ECTS programme | GPA: 7.6 | Supervisor: Yiannis Kamarianakis.
University of Crete: Department of Mathematics and Applied Mathematics.
Graduation requirements fulfilled in 9/2022, official graduation ceremony held in 7/2023.

Experience

- **Data Scientist | Toyota Motor Europe | Intern, Connected Powertrain (R&D) | Brussels, Belgium** (Dec. 2024 - June 2025)
  - Data Science, Machine Learning & Data Analysis, Time Series Analysis
  - Python, Git, RTBD (Real Time Big Data)
  - System Control (e.g.: Extended Kalman Filter), CAN Signals, APIs.
  - Hybrid Models, Physical & Data Driven Models
  - BEV Energy Flow (Vehicle, Powertrain, Climate & Battery)
  - BEV Energy Consumption Prediction, Range Recommendation and Speed Forecast
- **Generative AI Engineer | FORTH-Hellas | IACM Lab, R&D | Heraklion, Crete, Greece** (Oct. 2024 - July 2025)
  - Python, Poetry, PyTorch, Git, Docker.
  - Large Language Models, ESM, AlphaFold.
  - Generative AI with Diffusion Models, GANs, VAEs.
  - Transformer Architectures, LLMs, RAG (Retrieval-Augmented Generation)
  - Deep Learning, Transfer Learning, Representation Learning.
  - Conditional Generation, Ancestral Sequence Reconstruction (ASR).
- **Statistical Analyst | FORTH-Hellas | Intern, R&D | Heraklion, Crete, Greece** (Dec. 2022 - May 2023)
  - Applied Statistics & Statistical Learning using R and Python.
  - Predictive Modeling, Penalized Adaptive LASSO Estimators.
  - Developed Python packages for automated model selection and hyperparameter tuning.
  - Focus on Model Interpretability, Regularization.
- **University Teaching Assistant | Graduate Teaching | Heraklion, Crete, Greece** (Sep. 2022 - June 2024)
  - Machine Learning (Postgraduate) | Python Computer Language (Fall 2023)
  - Introduction to Linear Algebra (Fall 2022)
  - Numerical Analysis (Spring 2024)

Publications

- **MSc. Thesis: Generative AI in Protein Engineering using Large Language Diffusion Models** 2025
- **15-Minute Ahead Traffic Volume Forecasting in Athens using AR-Distributed Lag and GARCH Models with Robust Quantile Regression for Forecast Combination.** 2024
- **BSc. thesis: Supervised Classification with Parametric Models** 2023
  Supervisor: Yiannis Kamarianakis
- **Identification of Normal Modes in Underwater Acoustic Propagation using Convolutional Neural Networks.** 2022
  In Proceedings of 24th international congress on acoustics, ICA, Acoustical society, **Korea**, 2022. **Authors**: Costas Smaragdakis, John Maris, Michael Taroudakis.

Programming & Frameworks

Python
R
PostgreSQL
Mojo

scikit-learn, pandas, PyTorch, TensorFlow, statsmodels, tsRNN
seaborn, matplotlib, SymPy, SciPy, NumPy, forecast
cdt, MASS, caret, ggplot2, glmnet, quantreg